

Geometry of Particle Methods

Analogous formulas hold for $M_+(\mathcal{M})$ (positive msrs. on $\mathcal{M}$)!

Geometry of Splats

Definition (Bures-Wasserstein Manifold)

Fix an isotropic, centered density $\rho(\cdot) \in \mathcal{P}_2(\mathbb{R}^d)$ and let $\text{BW}_\rho(\mathbb{R}^d)$ contain all *affine pushforwards*,

$$\text{BW}_\rho(\mathbb{R}^d) = \{(A(\cdot) + b)_\# \rho : b \in \mathbb{R}^d, A \in \mathbb{R}^{d \times d}\}$$

endowed with the Wasserstein metric. This manifold is *geodesically convex* and *invariant to choice of ρ* .

“Splats” are elements of $\text{BW}_\rho(\mathbb{R}^d)$ and ρ is the “mother splat.”